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Occurrence and distribution of per- and polyfluoroalkyl substances (PFASs) in human livers with liver cancer

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ABSTRACT

Per- and polyfluoroalkyl substances (PFASs) are anthropogenic compounds that are widely accumulated in human tissues, and the liver is considered a primary target organ for PFASs exposure. The occurrence and distribution of 21 PFASs in liver tissues with tumors (n = 55) and without tumors (n = 55) are investigated in this study. Eleven perfluorinated carboxylic acids (PFCAs) and five perfluorinated sulfonic acids (PFSAs) were detected at high frequencies (45.5%–100 %), while the detection frequencies of five perfluoroalkyl phosphate (PFPAs) were relatively lower (≤29.1 %). PFSAs and PFCAs accounted for up to 82.5%–92.7 % of the total PFASs. Although it was not found to be statistically significant, the concentrations of the total PFASs were slightly higher in the tumor liver samples (mean 64.3, range 5.70–303 ng/g) than those in the non-tumor liver samples (mean 62.7, range 4.08–240 ng/g).The perfluorooctanoic acid (PFOA), perfluorotridecanoic acid (PFTrDA), and perfluorobutanesulphonate (PFBS) showed significant differences ($p < 0.05$) between the tumor and non-tumor liver samples, and the different distribution levels of these three PFASs may have been a consequence of oxidative stress. The total concentrations of PFASs in the three age groups were in the decreasing order of middle-aged people (45–60) > old people (>60) > young people (<45). The PFASs in females were generally lower than in males, which may have been related to women's special excretion methods (such as childbirth and breastfeeding). The results should be valuable for further mechanistic studies regarding the toxic effects of PFASs in human livers.

1. Introduction

Per- and polyfluoroalkyl substances (PFASs) are a group of anthropogenic fluorinated organic chemicals. Due to their unique thermal and chemical stability, low vapor pressure, and high solubility, PFASs have been widely used in household and industrial products as surface protectors in carpets, leather, paper, semiconductors, and plastics (Agency, 2000; Hekster et al., 2003). Most PFASs are extremely resistant to biodegradation in the environment due to the strong C–F bonds (Loi et al., 2011). PFASs can be detected in various environmental media and even human bodies (Wang et al., 2018, 2021b; Ryu et al., 2021). It has been reported that PFASs in the human body may induce a variety of diseases (such as thyroid disease and immune disorders) (Chang et al., 2016; Kim et al., 2018). Due to their potential adverse effects on human health, the distribution of PFASs in human tissues has attracted increasing attention worldwide.

Humans can take in PFASs through multiple exposure routes, such as food consumption, drinking water, and inhalation (D'Hollander et al., 2010; Hsu et al., 2013). PFASs have been widely detected in various foods (such as meat, vegetables, milk, and eggs) (Chu et al., 2016; Zhang et al., 2021), drinking water (Ericson et al., 2008; Jiang et al., 2021), and air (Zhou et al., 2021). Different from the lipophilicity of other persistent organic pollutants, PFASs have protein-philic properties. Once they enter the human body, PFASs will preferentially stick to proteins (Domingo, 2012). Therefore, it is easy for them to accumulate in biological tissues and organs with high protein contents, such as the blood, liver, kidney, and gallbladder (Verreault et al., 2005). The occurrence and distribution of PFASs in the urine and blood have been reported in many studies (Kim et al., 2014a; Aimuzi et al., 2019; Thepaut et al., 2021). The concentrations of perfluorohexane sulfonate (PFHxS) were

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found to be higher in blood than in urine, and this may have been due to the higher protein affinity of PFHxS in blood than in urine (Ericson et al., 2007; Olsen and Zobel, 2007). Previous studies have identified strong positive correlations between serum and urine concentrations of PFASs (Zhang et al., 2013a; Worley et al., 2017). Meantime, the relationship between age, gender, region and PFASs concentrations were widely studied. And it was reported that positive statistically significant ($p < 0.01$) correlations between age and PFOA, PFOS, and PFHxS have been reported by Hoelzer et al. (2008). This positive finding contrast with a previous study that the estimated serum PFOA level decreased with age (35–39 years) (Pitter et al., 2020). In an Australian study, the concentrations of PFHpA and 8:2 fluorotelomer sulfonic acid (8:2 FTSA) in serum showed negative association with age, while PFHpS and PFOS concentrations increased with age (Eriksson et al., 2017). Thus, there lacks a consistent conclusion of age effect on PFASs in human. Studies have shown that levels of most PFASs were higher in males than in females (Ericson et al., 2007; Ji et al., 2012; Wan et al., 2013; Kim et al., 2014b). In addition, people in urban areas showed higher PFAS levels compared to rural areas (Harada et al., 2004; Guruge et al., 2005; Fu et al., 2014), which may have been attributed to different dietary habits and industrial and commercial levels (Zhang et al., 2013b). Perez et al. (2013) analyzed the distribution of 21 PFASs in autopsy tissues (brain, liver, lung, bone, and kidney), and found that the highest concentrations of perfluorooctanesulfonic acid (PFOS), perfluorooctadecanoic acid (PFODA), and perfluorohexyl ethanoic acid (FHEA) were found in the liver tissues. Furthermore, perfluorooctanoic acid (PFOA) levels in the liver samples from Catalan males were significantly higher than those from females (Ericson et al., 2007), and people in the different regions showed different PFAS levels in the liver (Olsen et al., 2003; Maestri et al., 2006). Yeung et al. (2013) analyzed the PFASs in the liver of patients with liver cancer and cirrhosis, and they found that the average concentrations of PFOS and PFHxS in the tumor liver tissues were 6.24 and 0.314 ng/g, respectively, which were higher than those in cirrhosis liver tissues, with mean values of 5.03 and 0.312 ng/g, respectively. Currently, there are limited studies on PFASs in liver cancer tissues.

As a metabolism and detoxification organ, the liver is a high-fat organ and suitable for the accumulation of persistent organic pollutants. The liver has been proven to be the major organ for PFAS accumulations (Butenhoff et al., 2006; Nakayama et al., 2008). It has been reported that subacute and sub chronic PFASs exposure can cause liver inflammation (Wang et al., 2021a). PFASs can destroy the antioxidant homeostasis and affect the lipid composition and lipid-related pathways, cause abnormal liver histology, and ultimately lead to liver damage (Wang et al., 2021a). Animal studies have shown that PFOS and PFOA can potentially cause adverse reactions in the liver (Kennedy et al., 2004). In vivo and in vitro rat liver cell tests showed that PFOA can impair the liver function of mice, by inducing differentiation 36 (CD36)-labeled cells apoptosis and CD36 protein expression and disrupting key hormones in the pancreas (Wu et al., 2017). It was proposed that PFASs may cause disturbances in the intestinal microbiome, which have been shown to be involved in the pathogenesis of liver disease (Qin et al., 2014). Dietary carcinogenicity studies in rats have investigated the carcinogenic potential of PFOA, and the results showed that PFOA could induce a tumor triad, including adenomas of the liver, which are similar to several PPARα agonists (Biegel et al., 2001; Steenland et al., 2010; Klaunig et al., 2012). PFASs in rodents increased proliferation and then resulted in the formation of preneoplastic cells or the induction of new focal lesions. The selective clonal growth of lesions leads to the formation of liver tumors in rats (Klaunig et al., 2012). These animal studies have enhanced the understanding on the associations between liver toxicities and PFAS exposures. For example, a recent study reported that serum PFAS concentrations were associated with liver biomarker levels, suggested that PFASs were hepatotoxic (Nian et al., 2019).This positive finding was consistent with earlier longitudinal study, which indicated that serum gamma glutamyl transpeptidase (GGT) levels were positively correlated with serum PFOA concentrations of workers in PFOA-related plant ($p = 0.02$) (Sakr et al., 2007). In summary, these results provide some evidences than PFASs may have adverse effects on human livers, and direct evidences on the association between PFASs and liver disease was not clear.

In this study, the levels and distribution of 21 PFASs in tumor liver (TL) and non-tumor liver (NL) samples of humans are first determined. Furthermore, the differences in PFASs in the TL and NL samples are compared to demonstrate the special characteristics of PFASs in human livers with tumors. Finally, the influence of gender and age on the distribution of PFASs in the TL and NL are investigated. The results provide effective basic information regarding the relationship between PFAS exposure and liver tumor disease.

2. Materials and methods

2.1. Chemicals and standards

Standard solutions containing 21 PFASs were purchased from Wellington Laboratories Inc (Guelph, ON, Canada) (>98 % purity). A total of 21 PFASs analytes were classified as 11 perfluorinated carboxylic acids (PFCAs) (namely, perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), PFOA, perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUdA), perfluorododecanoic acid (PFDoA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA)), five perfluorinated sulfonic acids (PFSAs) (namely, perfluorobutanesulphonate (PFBS), PFHxS, perfluoroheptane sulfonic acid (PFHpS), PFOS, perflurodecanesulphonate (PFDS)), and five perfluorinated phosphonic acids (PFPAs) (namely, perfluorohexylphosphonic acid (PFHxPA), perfluorooctylphosphonic acid (PFOPA), perfluoro decyl phosphonic acid (PFDPA), sodium (1H,1H,2H,2H-perfluorooctyl) phosphate (6:2diPAP), and sodium (1H,1H,2H,2H-perfluorodecyl) phosphate (8:2diPAP)). Sodium perfluoro-1-[1,2,3,4–$^{13}C_4$] octane sulfonate ($^{13}C_4$ -PFOS) (MPFOS) and perfluoro-n-[1,2–$^{13}C_2$] octanoic acid ($^{13}C_2$ PFOA) (MPFOA) were used as the internal standard and surrogate standard, respectively. High performance liquid chromatography (HPLC) grade methyl-tert-butyl ether (MTBE) and methanol (MeOH) were purchased from Fisher Scientific (USA). Tetrabutylammonium hydrogen sulfate (TBA) (>98 % purity) was purchased from Aladdin Biochemical Technology (Shanghai, China). The ammonia (30 %) was purchased from XiLong Scientific (Guangdong, China). The analytical grade sodium carbonate (Na_2CO_3, ≥99.5 %), sodium bicarbonate ($NaHCO_3$, ≥99.7 %), and ammonium acetate (CH_3COONH_4, ≥98 %) were purchased from Sigma-Aldrich (St. Louis, MO, USA). The ultra-pure water was produced using a Milli-Q purification apparatus (Millipore, Bedford, MA, France).

2.2. Sample preparation

During 2018–2020, 55 cases of liver tumor and corresponding non-tumor liver samples derived from liver cancer patients were obtained from the Third Affiliated Hospital of Sun Yat sen University. Informed consents were signed by all of the participants prior to this study, and this study was approved by the Medical Ethics Committee of the Third Affiliated Hospital of Sun Yat sen University. The demographic characteristics of the human donors are shown in the Supplementary Table S1. Participants selected in this research ranged in age from 24 to 84, and they were divided into three groups according to age: < 45 years old, 45–60 years old, and >60 years old. According to the World Health Organization's classification of age groups: under 45 year olds are called young people, 45–60 year olds are called middle-aged people, and people over 60 year olds can be called old people ((WHO 2010)). The liver tissues were obtained immediately after liver resection and placed into Eppendorf tubes and stored in dry ice temporarily. All of the samples were transferred and stored in a refrigerator at −80 °C.

2.3. Pre-treatment of the liver samples

The liver samples were extracted using an ion-pair liquid-liquid extraction method based on a previous protocol (Hansen et al., 2001) with minor modifications. The collected liver samples were wrapped in aluminum foil for freeze-drying. Briefly, 0.1 g of each liver sample (dry weight) was weighed and spiked with 2.5 ng of labeled surrogate standard (MPFOA) (25 μL, 100 ng/mL) in a 15 mL polypropylene (PP) tube. The fortified sample was mixed with 1 mL 0.5 M TBA solution, 2 mL of 0.25 M $NaHCO_3/Na_2CO_3$ buffer solution (pH = 10), and then vortex–mixed thoroughly. Four milliliters of MTBE was added to the PP tube, and the sample mixtures were shaken for 30 min at 200 rpm/min. The organic and aqueous layers were separated by centrifugation at 7000 rpm for 10 min. Then the supernatant organic phase (approximately 3.5 mL) was removed and placed in another clean PP tube. The extraction process was repeated three times using 4 mL of MTBE. All of the rinses were combined and mixed thoroughly and concentrated to near dryness under a nitrogen gas flow at ambient temperature. After that, the concentrated solution was reconstituted in a mixture of 1 mL MeOH and 4 mL ultra-pure water.

The subsequent purification of the liver samples was performed using a solid phase extraction (SPE) clean-up step using an Oasis WAX (60 mg/3 mL) cartridge (Waters Corporation, Milford, MA, USA). The Oasis WAX SPE cartridges were conditioned with 4 mL 0.1 % NH_4OH in methanol followed by 4 mL MeOH and 4 mL ultra-pure water. The extract was loaded onto the cartridge. Then the inner wall of the PP tube was rinsed with 2 mL ultrapure water for three times, and the rinse solution was also transferred to the cartridge. After loading the rinse solution, the cartridges were washed with 4 mL 25 mM ammonium acetate (pH = 4) and dried in vacuum for 45 min. The target analytes were eluted to a new PP tube with 4 mL MeOH followed by 4 mL 0.1 % NH_4OH in the MeOH. The eluent was concentrated to near dryness before being reconstituted in methanol. Then the sample was filtered through a 0.22 μm inorganic membrane filter and transferred into an amber auto-sampler vial. Finally, 2.5 ng MPFOS (25 μL, 100 ng/mL) was spiked into the extracts as the injection standard before being analyzed by liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS).

2.4. Instrumental analysis

The analyses of the 21 PFASs in the liver samples were performed using LC-MS/MS (AB SCIEX TRIPLE QUAD 5500) in a negative pattern with negative electrospray ionization (ESI) source and a multiple reaction monitoring (MRM) mode. The separation of the PFASs was performed on a 100 mm × 2.1 mm Waters BEH C18 column with a 1.7 mm particle size. The initial gradient composition began at 20 % methanol, held for 1 min, linearly increased to 90 % methanol within 4 min, and maintained for 2 min. At 6.1 min, the ramp was decreased to 10 % methanol and maintained at this state until 8 min. The total run time was 8.1 min. The instrumental MS parameters settings for all of the target compounds are presented in the Supplementary Table S2.

2.5. Quality assurance and quality control

The concentrations of the PFASs in the liver samples were quantitatively analyzed using the internal-standard method. Eight point calibration curves with a series of concentration levels (0.5, 1, 2, 5, 10, 20, 50, and 100 ng/L) were used for quantification of the PFASs concentration. The correlation coefficients (R^2) for all the target compounds were higher than 0.99. The concentrations in the standards where the measured values showed more than a ±20 % deviation from the theoretical concentration were discarded. During the experiment, all of the experimental utensils were disposable, and they were pre-cleaned with methanol three times to reduce the background pollution. A solvent blank, procedure blank, and matrix spiked were run after every 10 samples to monitor the experimental background concentration and extraction efficiency. The spiking experiments were performed using blank animal (pig) matrices of the liver at three different concentration levels (5, 25, and 50 ng/g of tissue). The mean spike recovery was satisfactory for most of the PFASs, which was in the range of 73.3 %–101.4 %, with the exception of PFBA, PFPeA, PFOPA, PFDPA, 6:2diPAP and 8:2diPAP (21.4%–58.0 %). Recoveries of the labeled surrogate standards in the liver samples ranged between 94.2 % and 136.2 %. A mid-point calibration standard was injected after every 10 samples to monitor the instrumental stability. Methanol was injected after every batch of 10 samples as a check for the target analyte residuals between the samples. The limit of detection (LOD) was determined as the lowest point of the calibration curve with a signal-to-noise ratio of three (peak-to-peak algorithm), the limit of quantification (LOQ) was defined as a concentration with a signal-to-noise ratio of 10.The method quantification limit (MQL) was determined at the lowest concentration for analyzed matrix (a signal-to-noise ratio (S/N) ≥ 10) (Details shown in Table 1).The sample concentrations were reported as they underwent blank correction and recovery correction.

2.6. Statistical analysis

The data analysis was performed using the Statistical Package for the Social Sciences (SPSS), Version 26.0. Based on the quality of the extracted liver, unit conversions of the mass/volume to mass/mass were made. Concentrations less than the LOD were set as the LOD divided by $\sqrt{2}$ for further analyses. All of the statistical tests that required the normality hypothesis were performed under the natural logarithmic transformation. Six PFASs with detection frequencies below 50 % were excluded from the statistical analysis, including: PFTeDA, PFHxPA, PFOPA, PFDPA, 6:2 diPAP, and 8:2 diPAP. The Mann–Whitney's *U* test was used to clarify the difference in the distribution of the PFASs in the TL and NL samples. The Spearman rank correlation coefficients were computed to estimate the correlations among the pairs of PFAS concentrations in the liver samples. The nonparametric Kruskal-Wallis test and Mann–Whitney's U were used to assess the PFASs differences between males and females and among the three age groups, respectively. A $P < 0.05$ was defined as being statistically significant.

3. Results and discussion

3.1. PFAS concentrations and correlations in the liver samples

The concentrations of the 21 PFASs in the paired liver samples (TL and NL) from the same patient are summarized and are shown in Table 1. The detection frequencies and concentrations of the PFASs displayed slight variations between the TL and NL samples. Among the 21 PFASs, PFHxA, PFNA, PFUdA, PFHxS, PFHpS, and PFOS showed higher detected frequencies (greater than 92.7 %) than the other PFASs (Table 1). The detection frequencies of all PFASs in the NL samples were comparable with that in the TL samples, except for PFOA, PFTrDA, and PFTeDA. The detection frequencies of PFOA, PFTrDA, and PFTeDA in the NL samples were 54.5 %, 54.5 %, and 23.6 %, respectively, which were much lower than those in the TL samples (92.7 %, 87.3 %, and 45.5 %, respectively). Eleven PFCAs and five PFSAs were detected with high frequencies (45.5%–100 %) (Table 1), and they were the primary pollutants in both the TL and NL samples and accounted for up to 82.5%–92.7 % of the total PFASs. The detection frequency of the five PFPAs were lower (≤29.1 %) (Table 1), and the PFPA levels in the same sample appeared to be lower than the PFSAs and PFCAs. Since the lower global historical use and release of PFPA materials (Wang et al., 2016), they are often less frequently detected and quantified at lower levels than PFCAs and PFSAs (Guo et al., 2012; Ullah et al., 2012). The different distribution levels of the PFASs in the liver samples may have been caused by complicated reasons, such as the different exposure

Table 1
The concentrations (ng/g dry weight) of per- and polyfluoroalkyl substances (PFASs) in human liver tissues with tumors and without tumors.

Target Compounds	LOD	LOQ	MQL	Non-tumor liver Mean	Median	Range	DF	Tumor liver Mean	Median	Range	DF
PFBA	0.0300	0.100	0.990	8.43	4.21	<LOD-79.5	72.7 %	6.55	2.59	<LOD-101	76.4 %
PFPeA	0.150	0.490	0.519	2.04	1.01	<LOD-11.2	76.4 %	1.93	1.09	<LOD-25.5	69.1 %
PFHxA	0.610	2.04	2.28	2.65	1.92	<LOD-15.2	96.4 %	2.46	1.50	<LOD-15.6	94.5 %
PFHpA	0.390	1.30	1.94	2.59	1.73	<LOD-18.7	81.8 %	2.18	0.600	<LOD-17.2	60.0 %
PFOA	1.18	3.93	1.72	1.18	0.120	<LOD-13.3	54.5 %	3.19	1.92	<LOD-24.6	92.7 %
PFNA	0.980	3.26	0.379	1.10	0.870	<LOD-4.14	96.4 %	1.23	0.78	<LOD-13.1	98.2 %
PFDA	0.520	1.72	0.418	0.900	0.390	<LOD-6.02	65.5 %	0.73	0	<LOD-10.9	47.3 %
PFUdA	0.430	1.43	0.812	1.56	0.950	<LOD-8.80	92.7 %	1.26	0.620	<LOD-10.8	92.7 %
PFDoA	0.390	1.29	1.29	0.470	0.120	<LOD-6.97	52.7 %	0.300	0.110	<LOD-1.93	50.9 %
PFTrDA	0.370	1.24	2.09	0.530	0.120	<LOD-6.27	54.5 %	1.02	0.470	<LOD-6.40	87.3 %
PFTeDA	0.930	3.09	4.80	0.470	0	<LOD-5.03	23.6 %	0.170	0	<LOD-4.50	45.5 %
PFBS	0.100	0.340	3.48	0.570	0.130	<LOD-7.97	58.2 %	1.55	0.930	<LOD-7.95	96.4 %
PFHxs	0.480	1.60	2.92	2.92	2.46	<LOD-11.5	92.7 %	2.63	1.73	<LOD-18.0	92.7 %
PFHpS	0.370	1.25	2.00	6.13	3.60	<LOD-23.4	92.7 %	6.37	3.16	0.140–55.7	100 %
PFOS	0.150	0.490	2.86	22.2	15.6	<LOD-101	98.2 %	22.4	13.4	<LOD-157	96.4 %
PFDS	0.0600	0.210	1.12	4.35	0.0400	<LOD-52.8	56.4 %	0.960	0.280	<LOD-18.0	78.2 %
PFHxPA	1.08	3.60	1.04	0.620	0	<LOD-7.68	23.6 %	1.68	0	<LOD-24.6	20.0 %
PFOPA	3.19	10.6	0.745	2.28	0	<LOD-36.9	14.5 %	6.61	0	<LOD-75.8	18.2 %
PFDPA	1.49	4.95	0.216	1.45	0	<LOD-34.8	12.7 %	0.650	0	<LOD-19.1	5.50 %
6:2diPAP	4.62	15.4	5.20	0.200	0	<LOD-3.84	14.5 %	0.160	0	<LOD-1.90	29.1 %
8:2diPAP	1.05	3.51	1.54	0.0300	0	<LOD-0.960	3.60 %	0.210	0	<LOD-5.28	16.4 %
∑11PFCAs[a]	–	–		21.9	16.5	1.32–144	100 %	21.0	13.6	1.39–154	100 %
∑5PFSAs[b]	–	–		36.2	31.0	1.70–112	100 %	34.0	23.2	4.01–166	100 %
∑5PFPAs[c]	–	–		4.57	0	0–59.1	100 %	9.31	0	0–103	100 %
∑21PFASs[d]	–	–		62.7	48.0	4.08–240	100 %	64.3	41.7	5.70–303	100 %

AbbreviationsLOD, limit of detection; LOQ, limit of quantification; MQL, method quantification limit (S/N = 10), transformed to ng/g using the sample amount of the liver matrices; DF, detection frequency.
[a] The total concentration of 11 PFCAs.
[b] The total concentration of 5 PFSAs.
[c] The total concentration of 5 PFPAs.
[d] The total concentration of 21 PFASs.

environments of the different patients, the different historical releases, and the different pharmacokinetic behaviors and metabolic fates of the different PFASs in the human body (Lee and Mabury, 2011).

The total concentrations of the 21 PFASs in the liver of each participant are presented in Supplementary Fig. S1. The range of the ∑21 PFASs in the TL and NL samples was between 5.70 and 303 ng/g and 4.08–240 ng/g, which was generally lower than total PFAS levels in liver tissues from Spain (372.9 ng/g) (Perez et al., 2013). Such a difference may have been due to the difference in the geographical and living eating habits of the different sampling groups (Guo et al., 2011). In all of the liver samples, PFOS was detected with the highest concentration. This result was similar to a previous study on PFAS levels in cancerous livers in patients from Australia in which PFOS showed the highest concentration among all of the detected PFASs (Yeung et al., 2013). PFOS and PFHpS were the two primary PFASs observed in the human livers in this study, and this result may have been due to the fact that PFOS and PFHpS are commonly used in various consumer products (such as surfactant, textiles, upholstery, paper, and food packaging) (Khansari et al., 2017).

The Spearman rank correlation coefficients among the concentrations of the different PFASs in the liver samples are showen in the Supplementary Table S3. The individual PFAS levels in the liver samples were positively correlated with each other, and most of the correlations were highly significant ($p < 0.001$) (Fig. 1). It is interesting to note that significant positive correlations were observed between the PFSAs (including PFHpS, PFOS, and PFDS) and PFCAs (including PFBA, PFPeA, PFHxA, PFHpA, PFOA, and PFNA), which suggested common sources of human exposure to these PFASs in daily life (Yeung et al., 2013). In addition, the long-chain PFCAs were positive correlated the with short-chain PFCAs moderately. There are two possible reasons for such findings: (1) the historical joint emissions of short chain PFCAs with long chain PFCAs (the short chain PFCAs are primarily regarded as unintended by-products during the manufacturing of long chain PFASs) (Cousins et al., 2011); and (2) long chain carboxylate PFASs can

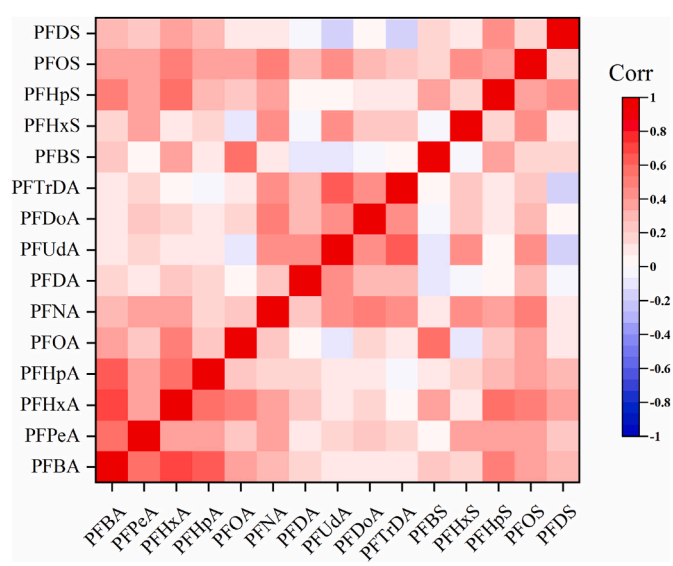

Fig. 1. The heat map of correlations between different PFASs.

biologically degrade to short chain PFASs (Martin et al., 2005; Nilsson et al., 2010). PFOS was positively correlated with PFOA ($\rho = 0.337$, $p < 0.001$), and similar results were also reported by Yeung et al. (2013) and Fu et al. (2014). As PFOS is difficult to degrade or metabolize to PFOA in humans (Kannan et al., 2004), the correlation between PFOS and PFOA may suggest common sources for human exposure to these two PFASs in daily life. Another reason may be that some PFOS precursors may degrade into PFOA due to environmental degradation (Yeung et al., 2013). Moreover, a significant positive correlation between the PFOS and PFHxS concentrations was observed. Since PFHxS is an impurity in

perfluorooctanesulfonyl fluoride- (POSF) based products, POSF and certain POSF-based products may eventually degrade into other fluorinated compounds during their life cycle, such as PFOS (Giesy and Kannan, 2001).

3.2. The difference in the PFAS distributions in tumor and non-tumor liver tissues

The Mann–Whitney's *U* test was performed to investigate whether there was significant difference in the PFAS distributions between tumor and non-tumor liver samples (Supplementary Table S4). The test results showed that no significant differences ($p > 0.05$) were observed between tumor and non-tumor liver samples (Fig. 2). The reason for the lack of distribution differences in the PFAS concentrations between tumor and non-tumor liver tissues may have been due to the limitations of the small sample size and the sample collection area. The average concentration of the 21 PFASs in the TL samples was 64.3 ng/g, higher than that in the NL samples (62.7 ng/g) (Table 1). The hepatic concentrations of most of the PFASs did not vary significantly between the tumor and non-tumor liver samples, except for PFOA, PFTrDA, and PFBS ($p < 0.05$) (Fig. 2). The tumor livers had significantly higher PFTrDA concentrations (mean 1.02 ng/g) than that in the non-tumor liver samples (mean 0.534 ng/g). In the TL samples, the average concentrations of PFOA and PFBS were 3.19 ng/g and 1.55 ng/g, respectively, which were 2.7 times ($p < 0.001$) higher than those in the NL samples.

PFASs can be excreted from the body through urine and kidneys (Harada et al., 2005; Cui, 2010), while most PFASs have shown a s low elimination rate in the human body (Jian et al., 2018). The liver is considered the primary target organ for PFAS exposure, and most liver cancers are closely linked to metabolic conditions (Yeung et al., 2013). Oxidative stress associated with the antioxidant capacity in the liver may be the reason for the different distributions of PFOA, PFTrDA, and PFBS observed in the TL and NL samples (Fig. 2). Previous reports in patients with hepatomas showed decreased ascorbic acid levels in the plasma, increased hepatic levels of malondialdehyde (MDA), and reduced levels of glutathione than in normal subjects (De Maria et al., 1996; Yamamoto et al., 1998). This indicated patients with liver cancer were subject to oxidative stress. Exposure to organic pollutants may cause oxidative stress (Kannan et al., 2006). PFASs have been proven to be pro-oxidants, and exposure to PFASs elicit the initiation of lipid peroxidation, the generation of reactive oxygen species (ROS), and the inhibition of antioxidant enzymes (Zeeshan et al., 2020). For example,

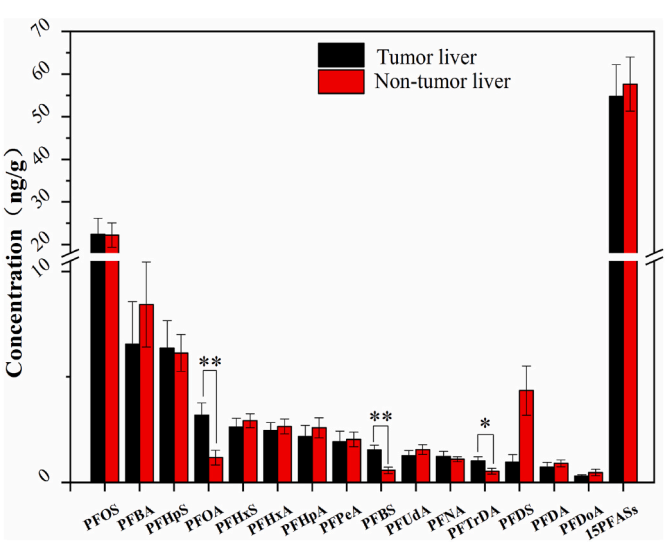

Fig. 2. Concentration (mean ± SE) of detected fifteen PFASs in the human liver tissues with and without tumor. The asterisk (*) indicates significant differences.

Liu et al. (2007) used primary cultured tilapia hepatocytes and reported that PFOA and PFOS could produce oxidative stress and induce cell apoptosis. Furthermore, PFASs can evoke lipid peroxidation and reduce cell viability (Slotkin et al., 2008). Yeung et al. (2013) suggested that pathological changes in the liver may alter the distribution of PFAS compounds in diseased livers. However, studies regarding PFASs in the liver system are limited, and further studies are required to evaluate the influence of liver pathological changes on PFAS distributions in the liver.

3.3. Age and gender effects on the PFAS distributions

Table 2 summarizes the mean concentrations of the 15 detected PFASs in all of the liver samples from the different age ranges. No obvious statistically significant difference in the ages was observed in the total PFASs concentrations of the liver samples in this study ($p > 0.05$) (Table 2). This result was similar to a previous study on serum levels, and these results showed that no data showed a significant relationship between age and PFASs levels in serum (Olsen et al., 2003; Calafat et al., 2007). Due to the limitations of the sample numbers and geographic differences, these results do not totally reflect the age-related accumulation of PFASs in humans. However, there were also some interesting findings observed in this study. For example, the average concentrations of the predominant PFASs (including PFOS, PFBA, and PFHpS) were higher in the 45–60 age group than in the other groups. The total concentrations of PFASs in the different age groups followed a decreasing order: middle-aged people (45–60) > old people (>60) > young people (<45), with average concentrations of 54.90, 50.44, and 48.81 ng/g, respectively (Table 2). Although Haug et al. (2009) revealed that most PFAS concentrations in human serum samples increased with age, the exposure of PFASs should be more important for the distribution of PFASs in human tissues. In the human body, PFASs can undergo biliary-enterohepatic circulation. Extensive hepatointestinal circulation of PFASs could prolong their biological half-life, which may explain the lack of an observed general tendency of increasing PFAS concentrations with age (Calafat et al., 2006).

For all of the donors (24–84 years), statistically significant differences ($p < 0.05$) between age and concentrations of PFHxS were found, while no significant association was found in the present study between age and the other PFASs (Table 2). A negative increase in PFHxS ($r = -0.153$, $p > 0.05$) concentrations with age was found in the NL samples, as shown in Fig. 3. The highest concentration of PFHxS was found in people younger than 45 years old (mean 3.86 ng/g) (Fig. 3). Similarly, it was observed that the PFHxS concentration in the serum was significantly higher in the 25 ± 5 years old age group (4.65 ± 3.7 ng/mL) than in 55 ± 5 age group (2.48 ± 1.39 ng/mL) (Ericson et al., 2007). Being exposed to the surface of carpet or upholstery treated with formulations containing PFHxS may be an important contributing factor for the higher PFHxS concentrations in young people (<45 years old) (Sundstrom et al., 2012). In contrast, Zhang et al. (2010b) found a significant increase in the PFHxS ($r = 0.357$, $p < 0.01$) concentration with age in blood. A possible reason for the significant positive correlation between blood PFHxS concentration with age may be related to dietary intake, especially fish and seafood, which have been reported to be the primary sources of PFHxS exposure (88 %) in the daily intake of PFASs of adults in China (Ericson et al., 2007; Zhang et al., 2010a). However, Kannan et al. (2004) and Yeung et al. (2006) revealed that there was no association between age and PFHxS concentrations. The inconsistent results of the age effect on PFASs in human tissues may be attributed to confounding factors, such as dietary exposures, occupation, and some physiological factors (Karrman et al., 2006; Schecter et al., 2012).

The influence of gender on the concentrations of PFASs was also examined, and the results are shown in Table 2. There were no obvious significant statistically differences ($p > 0.05$) in the total PFAS concentrations in the liver samples between the two gender groups. The lack of gender-related accumulation patterns of PFASs in the liver samples

Table 2
Mean concentrations of 15 detected PFASs in different groups with liver cancer (grouped by age and gender).

Non-tumor liver	PFBA	PFPeA	PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUdA	PFDoA	PFTrDA	PFBS	PFHxs	PFHpS	PFOS	PFDS	∑15PFASs
<45	4.49	1.08	1.60	1.35	0.910	0.940	1.10	1.97	0.350	0.570	0.630	3.86	4.24	26.1	1.43	49.3
45–60	10.0	2.83	3.15	3.07	1.75	1.14	0.960	0.960	0.790	0.450	0.540	2.49	8.54	17.8	7.28	60.5
>60	8.83	1.28	2.85	2.97	2.22	1.33	1.02	2.05	0.490	1.00	0.740	2.64	3.91	20.4	1.86	52.3
p-value	0.66	0.14	0.45	0.44	0.40	0.91	0.62	0.26	0.80	0.62	0.75	0.05	0.12	0.85	0.41	0.38
male	8.99	2.05	2.78	2.87	1.81	1.15	1.01	1.51	0.570	0.690	0.630	2.92	6.17	19.8	4.64	57.6
female	5.99	1.87	2.52	1.64	1.15	1.16	0.980	1.28	0.870	0.320	0.540	2.25	7.25	23.0	3.93	54.7
p-value	0.86	0.90	0.25	0.78	1.00	0.97	0.96	0.42	0.56	0.05	0.04	0.46	0.18	0.37	0.13	0.47
Tumor liverr																
<45	3.54	0.970	1.98	2.62	1.85	0.920	1.98	1.74	0.510	1.39	1.36	2.13	5.01	22.6	0.420	48.3
45–60	6.69	1.59	2.72	1.36	4.17	1.49	0.620	0.790	0.410	0.650	1.47	2.47	6.85	23.5	0.720	54.6
>60	6.66	1.80	2.03	3.08	2.67	1.17	0.830	1.90	0.460	1.53	1.43	2.83	4.62	17.5	0.760	48.6
p-value	0.58	0.97	0.65	0.45	0.94	0.62	0.07	0.79	0.83	0.93	0.98	0.28	0.94	0.94	0.71	0.73
male	6.80	1.58	2.56	2.33	3.43	1.33	0.990	1.21	0.440	1.10	1.34	2.61	6.05	20.7	0.740	52.2
female	1.70	1.14	1.31	0.86	2.32	1.03	0.83	1.81	0.460	0.780	2.00	1.85	4.66	26.7	0.280	47.8
p-value	0.31	0.72	0.28	0.15	0.98	0.58	0.19	0.63	0.98	0.80	0.38	0.71	0.41	0.66	0.31	0.57

*bold: Significance level is $p < 0.05$.

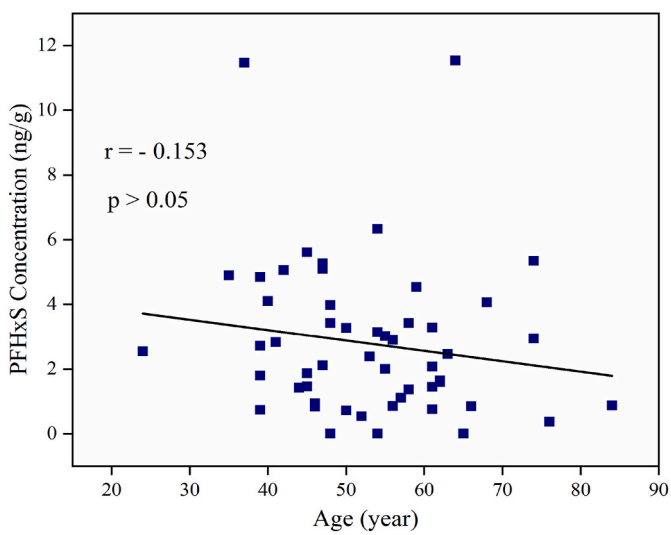

Fig. 3. The correlation between PFHxS concentrations in the NL samples and ages.

may have been related to the limitations of the sample scales and the affinity of PFASs for lipoproteins rather than neutral lipids (Luebker et al., 2002). However, the average concentration of the total PFASs in the liver samples of males (54.9 ng/g) was slightly higher than those in females (51.2 ng/g). Furthermore, the average concentrations in this study of most of the individual PFASs were higher in males than in females. A similar result was reported by Karrman et al. (2010), in which PFOA levels in the liver samples from Catalan males (0.96 ng/g) were significantly higher ($p < 0.05$) than those in females (0.45 ng/g). There was generally a significant difference ($p < 0.05$) between the sexes and the concentrations of PFASs. The PFTrDA and PFBS concentrations in men were 2.13 times and 1.18 times higher than that in women. Previous studies have shown most PFAS concentrations in human blood were higher in males than in females (Hsu et al., 2013; Kim et al., 2014b). Lower concentrations of the various PFASs in females than males might be related with women's specific excretion pathways, such as childbirth, breast milk and menstruation (Harada et al., 2005). In addition, animal experiments have revealed that female rats eliminated PFASs (including PFOA, PFHxS, PFNA and PFDA) more rapidly than male rats (Choi et al., 2020). Thus, the exposure of PFASs and pharmacokinetic differences of the different PFASs in human bodies may have been the primary reasons for the gender differences in the PFAS concentrations in the liver samples.

4. Conclusions

The occurrence and distribution of PFASs in human tisssues have been well documented, but little information has been reported regarded PFAS contamination levels in the human liver. The results of the present study showed that 21 PFASs could be detected in liver tissues from liver cancer patients. Among all of the PFASs, PFBA, PFHxS, PFHpS, and PFOS were found to be the predominant compounds. Significant positive correlations were noted in the concentrations of most PFASs, implying that these PFASs in the human liver may have a common source. The concentrations of PFOA, PFTrDA, and PFBS showed significant differences in tumor and non-tumor liver samples, and this may have been a consequence of oxidative stress. Furthermore, the participants aged 45–60 years showed higher PFASs concentrations in than the other aged people. Regarding gender, the liver levels of most PFASs were higher in males than in females. The PFAS exposure in the general population is ubiquitous, and further studies of the epidemiologic literature and the relationship between PFAS exposure and chronic diseases in humans are required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2021.111775.

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